$SrTiO_3$: Worked solutions

> Reference: Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

E01 Impedance time scale

For a parallel RC, admittance is $Y = R^{-1} + i\omega C$, hence $Z = R/(1 + i\omega RC)$. With $u = \omega RC$, $-\operatorname{Im} Z = Ru/(1 + u^2)$, whose maximum for $u > 0$ occurs at $u = 1$. Thus $\tau = RC = (10^6)(10^{-6}) = 1\,\mathrm{s}$ and $f_{peak} = 1/(2\pi\tau) = 0.159\,\mathrm{Hz}$. Other physical structures can have the same transfer function, so the peak is not a unique microscopic identification.

E02 Charged vacancies and slab resistance

For speed per field $v = \mu_V E$, each carrier carries charge magnitude $2q$, so $\sigma_V = 2qc_V\mu_V$. Substitution gives $2(1.602 \times 10^{-19})(10^{18})(8.6 \times 10^{-14}) = 2.755 \times 10^{-14}\,\mathrm{S/cm}$. The uniform slab gives $R = L/(\sigma_V A) = 0.005/(2.755 \times 10^{-14}) \approx 1.81 \times 10^{11}\,\Omega$. Other electronic carriers, nonuniform concentration, interfaces and uncertainty in mobility can invalidate this isolated bulk result.

E03 Contact barrier

Under the equal-prefactor and equal-effective-bias conditions, $I \propto e^{-\Phi_B/(k_BT)}$, giving $I_{0.50}/I_{0.72} = e^{(0.72-0.50)/0.02585} = e^{8.51} \approx 4.97 \times 10^3$. Different ideality factors, contacts, barrier shape, series resistance, temperatures or injection mechanisms change the ratio. In particular, it cannot be presented as a measured device switching ratio when the assumptions do not match the experiment.

E04 Two terminal voltages

Subtract the signed terminal voltages: $+2 - (-1) = +3\,\mathrm{V}$; $+2 - (+2) = 0\,\mathrm{V}$. An update can also depend on pulse width, time history, threshold distribution and temperature; peak voltage alone does not determine it.

E05 A finite device sample

The observed fraction is $16/40 = 0.40 = 40\%$. It is conditional on the tested batch, contact, protocol and the criterion; finite-sample uncertainty and selection effects also matter. It cannot establish a universal yield or predict another process without validation.

E06 Pulse-count reduction

The fractional reduction is $(62 - 16)/62 = 46/62 \approx 0.7419$, or 74.2%. It also describes pulse-train duration reduction only under a timing convention with equal pulse widths and spacings and the same stopping criterion; initial delays and off-by-one differences in interval count must be handled consistently.

E07 Tracer diffusion and conductivity

$D_\sigma = D^*/H = 6.9 \times 10^{-12}/0.69 = 1.0 \times 10^{-11}\,\mathrm{cm^2/s}$. Then

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$$\sigma = \frac{(1.0 \times 10^{-11})(10^{18})(2 \times 1.602 \times 10^{-19})^2}{(1.381 \times 10^{-23})(700)} \approx 1.06 \times 10^{-10}\,\mathrm{S/cm}.$$

The carrier concentration and Nernst–Einstein relation are model inputs, not automatic outputs of the tracer profile. Correlation, nonideal activity and interfaces can change interpretation; fitting a profile does not uniquely identify every microscopic mechanism.

E08 Sweep time

$0.05\,\mathrm{mV/s} = 5 \times 10^{-5}\,\mathrm{V/s}$, so a 1 V sweep takes $1/(5 \times 10^{-5}) = 2.0 \times 10^{4}\,\mathrm{s}$. At $25\,\mathrm{V/s}$ it takes $1/25 = 0.04\,\mathrm{s}$. Compared with the assumed 1 s internal time, the first is much slower and the second faster. These are sweep durations, not measured switching times; other faster processes, a large voltage or heating can still disturb the state.

E09 Impedance and microscopic hypotheses

Compare contact materials and device areas at fixed history and geometry, film thickness scaling, temperature and frequency response, and spatial or chemical measurements before and after switching. Trapping and oxygen migration may both change an interface barrier: controlled time, temperature, oxygen activity and spectroscopy can distinguish their predictions but no single I–V loop uniquely decides between them. Controls such as reference contacts, instrument checks and repeat cycles constrain alternate explanations. A delayed internal-state contribution or an instrument artifact can produce loop-like behavior without a literal inductor. Retention requires a stated zero-bias wait and subsequent low-disturbance read, repeated over times and devices.

E10 Timing-dependent update

For example define $\Delta t = t_{post} - t_{pre}$, label which waveform goes to Ni and Ti, and calculate $V_{device}(t) = V_{Ni}(t) - V_{Ti}(t)$ across their complete overlap. Prepare a common baseline or explicitly model its history; read at a fixed low-disturbance condition and report the conductance change and chosen normalization. Otherwise differing starting states or denominators can create an apparent timing trend.

E11 Film thickness and impedance

With $R = d/(\sigma A)$ and $C = \epsilon A/d$, doubling d doubles R and halves C, leaving $RC = \epsilon/\sigma$ unchanged under this ideal uniform-film model. Contact resistance or capacitance, current crowding, defects and changing microstructure can produce deviations or mimic trends, so geometry alone does not exclude an interface.

E12 Activation-energy inference

Taking logarithms gives $\ln(\sigma T) = \ln C - (E_a/k_B)(1/T)$. The slope is $-E_a/k_B$; hence $E_a = (9.28 \times 10^3)(8.617 \times 10^{-5})\,\mathrm{eV} \approx 0.800\,\mathrm{eV}$. Multiple carriers, contact changes, temperature-dependent prefactors or a changing structure can alter the apparent slope; independent evidence is needed to identify a carrier and validate the fit range.

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E13 A check circuit for associative learning

With the output taken across R_s, the divider gives $V_{out} = V_{check} R_s/(R_s + R_{mem})$. At $R_{mem} = 90\,\mathrm{k\Omega}$, $V_{out} = 0.3(10/100) = 0.030\,\mathrm{V} = 30\,\mathrm{mV}$, below the criterion. At $50\,\mathrm{k\Omega}$, it is $0.3(10/60) = 0.050\,\mathrm{V} = 50\,\mathrm{mV}$, strictly above 45 mV. These resistances are hypothetical; a divider calculation does not specify the pulse-driven state evolution or the number of trials. Verify read disturbance experimentally and do not infer biological learning from the electrical threshold.

E14 Interpret a reported sequence

Each paired CS–US sequence is one training trial; CS precedes US by 4 ms. The separate check pulse reads the divided response against the specified 45 mV criterion. Three pairs reached the criterion in this reported setup. After 30 pairs, repeated CS-only pulses were extinction trials, and the reported response fell below the criterion after ten. To test read disturbance, compare matched devices or reset states subjected to many checks versus fewer checks, with identical wait times and low-disturbance reference reads; also vary check amplitude and width. These outcomes depend on the device, initial state, pulse timing and decision threshold, so they do not establish universal trial counts or prove biological learning.

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https://github.com/xinhi6716/resistive-memory-self-study